SOC SOAR Capstone Project – Malware Analysis and Detection on Windows Endpoint

Project Overview

This project demonstrates a **realistic end-to-end SOC detection workflow** focused on **malware execution and persistence techniques on a Windows system**.

To ensure safe, real-world simulation without risking a personal machine, the entire exercise was performed on an **AWS Windows EC2 instance**, emulating an enterprise endpoint monitored by a SOC.

The objective of **Part 1** is to:

- Simulate **malware activity** on a Windows host
- Detect and correlate malicious behavior using **Sysmon + Wazuh**
- Perform **threat hunting and log analysis**
- Validate malicious activity before initiating incident response actions

This part focuses strictly on **detection, analysis, and confirmation**.
Incident response, case handling, SOAR automation, and threat sharing are covered in Part 2.

Why This Project Matters

Modern SOC operations rely on:

- Endpoint telemetry (Sysmon)
- SIEM correlation (Wazuh)
- Analyst-driven threat hunting
- MITRE ATT&CK–aligned detection

This project replicates how a **SOC analyst investigates suspicious alerts**, correlates telemetry, and confirms a **true malware incident** before escalation.

Environment & Tools Used

Infrastructure

- **AWS EC2 – Windows Server Instance**
- Isolated cloud environment for safe malware simulation

Security & Monitoring Tools

- **Wazuh Agent –** Endpoint monitoring and SIEM ingestion
- **Sysmon** – High-fidelity Windows event telemetry
- **Wazuh Dashboard** – Alert correlation and threat hunting

Analyst Role Simulated

- SOC Analyst (Tier 1 → Tier 2)
- Threat Detection & Investigation
- Malware behavior analysis
- MITRE ATT&CK mapping

Project Architecture

1. **Windows EC2 Endpoint**
   - Runs Wazuh Agent and Sysmon
2. **Sysmon**
   - Captures detailed process, file, registry, and network activity
3. **Wazuh Manager**
   - Ingests Sysmon events
   - Applies detection rules
4. **Wazuh Dashboard**
   - Visualizes alerts
   - Enables threat hunting and correlation

5. **SOC Analyst**
   - Investigates alerts
   - Confirms malicious activity

Project Scope – Part 1

✓ Endpoint setup
✓ Sysmon deployment
✓ Malware simulation
✓ Alert generation
✓ Threat hunting
✓ Log correlation
✓ Malware confirmation

- **Covered in Part 2**
  - Incident response execution
  - Case management
  - SOAR automation
  - Threat sharing

Step-by-Step Execution (Part 1)

Step 1: Windows EC2 Instance Setup

A **Windows EC2 instance** was deployed on AWS to act as a corporate endpoint.

Purpose:

- Enable realistic malware testing
- Avoid risk to personal systems
- Simulate enterprise Windows behavior

The instance was configured with:

- Administrator access
- Internet connectivity
- Baseline Windows configuration

Step 2: Wazuh Agent Installation

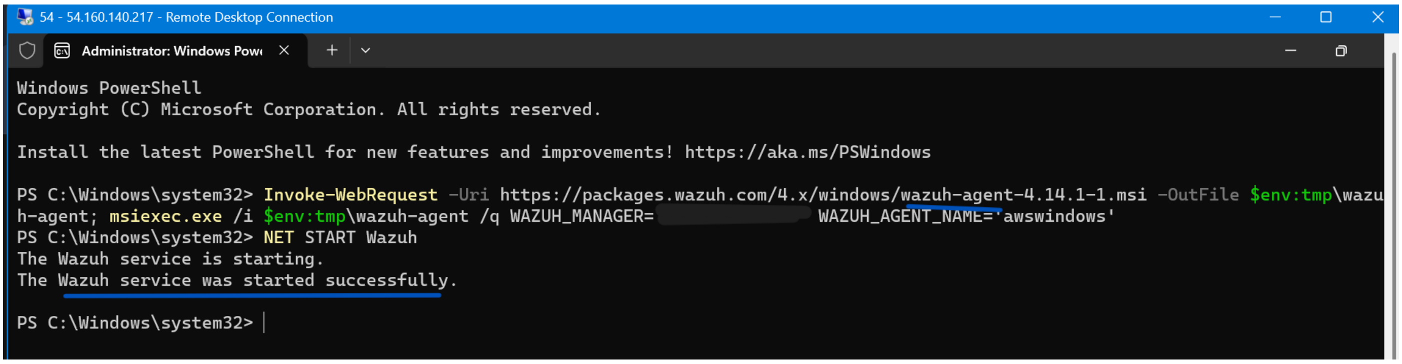
The **Wazuh agent** was installed on the Windows EC2 instance and successfully connected to the Wazuh manager.

Purpose:

- Collect endpoint logs
- Forward security telemetry to the SIEM
- Enable centralized detection

Outcome:

- Agent registered successfully
- Endpoint visible in Wazuh dashboard
- Log ingestion confirmed



Step 3: Sysmon Installation & Validation

Sysmon was installed with a production-grade configuration.

Purpose:

- Capture high-value security events such as:
  - Process creation
  - Image loads
  - File creation
  - Registry modifications
  - DNS queries

Validation Performed:

- Sysmon service running
- Sysmon operational event channel available
- Events flowing into Wazuh

```
System Monitor v15.15 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2024 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com

Loading configuration file with schema version 4.90
Configuration file validated.
Sysmon64 installed.
SysmonDrv installed.
Starting SysmonDrv.
SysmonDrv started.
Starting Sysmon64..
Sysmon64 started.
PS C:\Sysmon\Sysmon> Get-Service Sysmon64

Status   Name               DisplayName
------   ----               -----------
Running  Sysmon64           Sysmon64


PS C:\Sysmon\Sysmon> wevtutil el | findstr Sysmon
Microsoft-Windows-Sysmon/Operational
PS C:\Sysmon\Sysmon>
```

Step 4 – Malware Simulation: Suspicious File Execution Attempt

Description

An attempt was made to execute a suspicious .scr file masquerading as a legitimate Windows process (svchost). Execution failed with *Access is denied*, but the attempt itself is significant.

Why This Matters

.scr files are executable and commonly abused by malware. Attempted execution from a non-standard directory indicates malicious intent.

Step 5 – Malware Simulation: Repeated PowerShell Web Requests

Description

PowerShell was used in a loop to repeatedly send HTTP requests to an external domain, simulating automated communication or payload retrieval behavior.

Why This Matters

Repeated PowerShell web requests resemble command-and-control beaconing or staged malware download activity.

```
C:\Sysmon>C:\Users\Public\svchost-update\svchost_update.scr
Access is denied.

C:\Sysmon>for /L %i in (1,1,6) do powershell -c "Invoke-WebRequest http://testphp.vulnweb.com -UseBasicParsing"
```

Step 6 – Malware Simulation: PowerShell Execution and Registry Persistence

Commands Executed

powershell -c "Invoke-WebRequest http://testphp.vulnweb.com -UseBasicParsing"

reg add HKCU\Software\Microsoft\Windows\CurrentVersion\Run /v svchost_update /t REG_SZ /d "C:\Users\Public\svchost-update\svchost_update.scr" /f

Description

A PowerShell command successfully accessed an external site, followed by the creation of a registry Run key to establish persistence for the suspicious executable.

Why This Matters

Registry Run keys are a common persistence technique used by malware to ensure execution on user logon.

```
C:\Sysmon>powershell -c "Invoke-WebRequest http://testphp.vulnweb.com -UseBasicParsing"


StatusCode        : 200
StatusDescription : OK
Content           : <!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN"
                    "http://www.w3.org/TR/html4/loose.dtd">
                    <html><!-- InstanceBegin template="/Templates/main_dynamic_template.dwt.php" codeOutsideHTMLIsLoc...
RawContent        : HTTP/1.1 200 OK
                    Transfer-Encoding: chunked
                    Connection: keep-alive
                    Content-Type: text/html; charset=UTF-8
                    Date: Fri, 06 Feb 2026 19:53:09 GMT
                    Server: nginx/1.19.0
                    X-Powered-By: PHP/5.6.40-38+ubun...
Forms             :
Headers           : {[Transfer-Encoding, chunked], [Connection, keep-alive], [Content-Type, text/html; charset=UTF-8], [Date, Fri,
                    06 Feb 2026 19:53:09 GMT]...}
Images            : {}
InputFields       : {}
Links             : {@{outerHTML=<a href="https://www.acunetix.com/"><img src="images/logo.gif" width="306" height="38" border="0"
                    alt="Acunetix website security"></a>; tagName=A; href=https://www.acunetix.com/}, @{outerHTML=<a
                    href="https://www.acunetix.com/vulnerability-scanner/">Acunetix Web Vulnerability Scanner</a>; tagName=A;
                    href=https://www.acunetix.com/vulnerability-scanner/}, @{outerHTML=<a href="index.php">home</a>; tagName=A;
                    href=index.php}, @{outerHTML=<a href="categories.php">categories</a>; tagName=A; href=categories.php}...}
ParsedHtml        :
RawContentLength  : 4958
```

```
C:\Sysmon>
C:\Sysmon>reg add HKCU\Software\Microsoft\Windows\CurrentVersion\Run /v svchost_update /t REG_SZ /d "C:\Users\Public\svchost-update\s
vchost_update.scr" /f
The operation completed successfully.

C:\Sysmon>
```

Step 6: Initial Alert Detection in TheHive (SOC View)

The SOC analyst receives **multiple alerts**, including:

- Executable dropped in suspicious directory
- Registry value modification
- PowerShell execution activity

Analyst Actions:

- Reviewed alert metadata
- Checked severity and rule context
- Identified strong indicators of malware behavior

Status	Severity	Title	# Case	Type / Source / Reference	Details	Assignee	Dates
New New 23 minutes ago	M	Value added to registry key has Base64-like pattern rule=92041 agent_id=010 agent_name=awswindows agent_ip=10.0.1.7 wazuh None	-	wazuh_alert wazuh e0c06a	Observables 3 TTPs 0	?	O. 07/02/2026 01:23 C. 07/02/2026 01:23 U. 07/02/2026 01:23
New New 23 minutes ago	M	Executable file dropped in folder commonly used by malware agent_id=010 agent_name=awswindows rule=92213 agent_ip=10.0.1.7 wazuh None	-	wazuh_alert wazuh 4f14a9	Observables 1 TTPs 0	?	O. 07/02/2026 01:23 C. 07/02/2026 01:23 U. 07/02/2026 01:23
New New 23 minutes ago	M	Executable file dropped in folder commonly used by malware agent_id=010 agent_name=awswindows rule=92213 agent_ip=10.0.1.7 wazuh None	-	wazuh_alert wazuh bc213a	Observables 1 TTPs 0	?	O. 07/02/2026 01:23 C. 07/02/2026 01:23 U. 07/02/2026 01:23
New New 23 minutes ago	M	Executable file dropped in folder commonly used by malware agent_id=010 agent_name=awswindows rule=92213 agent_ip=10.0.1.7 wazuh None	-	wazuh_alert wazuh 75e3c4	Observables 1 TTPs 0	?	O. 07/02/2026 01:23 C. 07/02/2026 01:23 U. 07/02/2026 01:23

Step 7: Initial Investigation & Escalation

After reviewing alert details, the analyst:

- Marked alert as suspicious
- Added investigation comments
- Initiated deeper analysis in Wazuh

Decision:

Alerts require correlation and confirmation via SIEM logs.

Step 8: Threat Hunting via Wazuh Dashboard

Using the Wazuh threat hunting dashboard:

- Multiple high-severity alerts observed
- Rule groups associated with Sysmon
- Clear clustering of related events
- Six High Critical Alerts

Outcome:

- Evidence of coordinated malicious activity

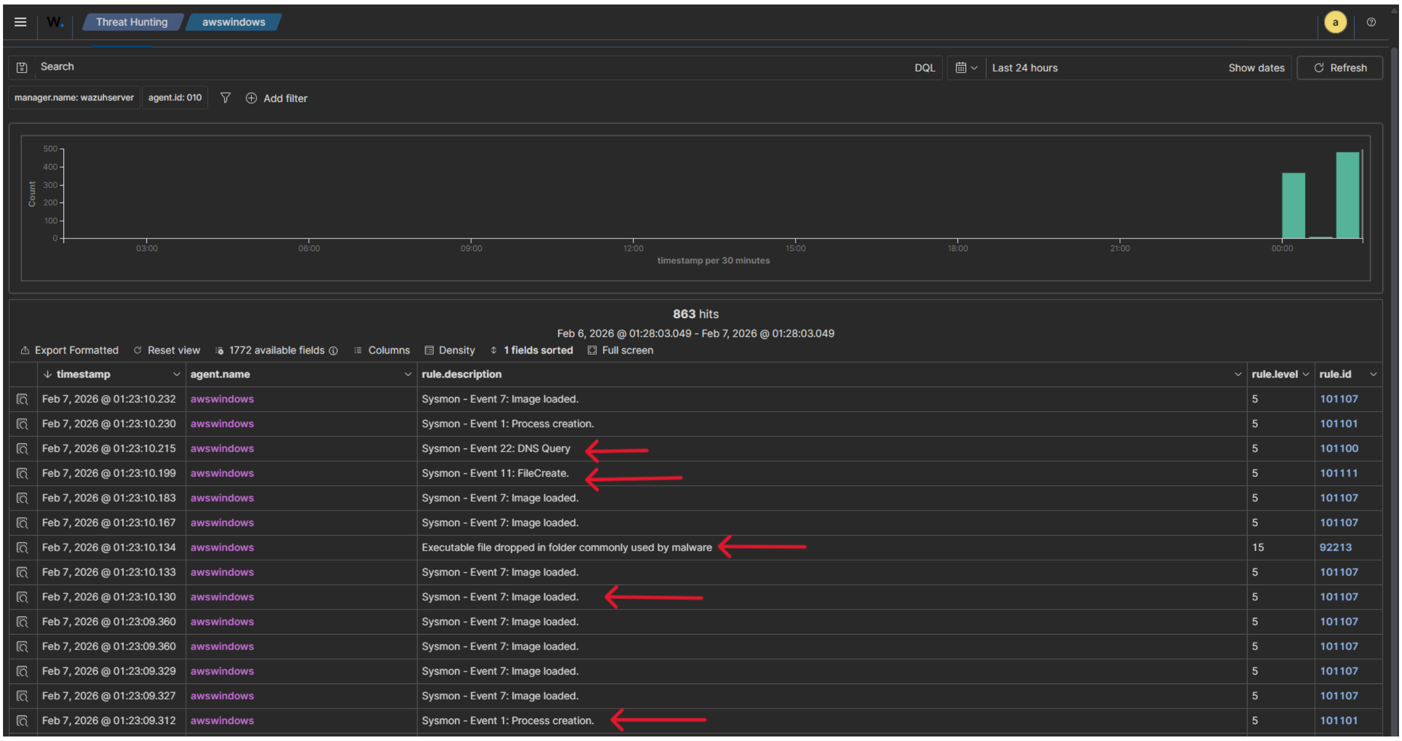
Step 9: Event Correlation Analysis

Multiple correlated alerts were identified:

- Process creation
- Image load
- File creation
- Registry modification
- DNS queries

Key Observation:

These events form a **single malware execution chain**, not isolated activity.



Step 10: Each Alert correlation detailed Analysis

Sysmon Event 1 – Process Creation

Analysis revealed:

- PowerShell execution command
- Suspicious command-line arguments
- MITRE technique mapping visible

MITRE Mapping:

- Execution via PowerShell

t	_index	wazuh-alerts-4.x-2026.02.06
t	agent.id	010
t	agent.ip	10.0.1.7
t	agent.name	awswindows
t	data.win.eventdata.commandLine	powershell -c \"Invoke-WebRequest http://testphp.vulnweb.com -UseBasicParsing\"
t	data.win.eventdata.company	Microsoft Corporation
t	data.win.eventdata.currentDirectory	C:\\Sysmon\\
t	data.win.eventdata.description	Windows PowerShell
t	data.win.eventdata.fileVersion	10.0.26100.5074 (WinBuild.160101.0800)
t	data.win.eventdata.hashes	SHA1=EB42621654E02FAF2DE940442B6DEB1A77864E5B,MD5=A97E6573B97B44C96122BFA543A82EA1,SHA256=0FF6F2C94BC7E2833A5F7E16DE1622E5DBA70396F31C7D5F56381870317E8C46,IMPHASH=AFACF6DC9041114B198160AAB4D0AE77
t	data.win.eventdata.image	C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe
t	data.win.eventdata.integrityLevel	High
t	data.win.eventdata.logonGuid	{03046e1c-3867-6986-2608-3e0000000000}
t	data.win.eventdata.logonId	0x3e0826
t	data.win.eventdata.originalFileName	PowerShell.EXE
t	data.win.eventdata.parentProcessGuid	{00000000-0000-0000-0000-000000000000}
t	data.win.eventdata.parentProcessId	1580
t	data.win.eventdata.processGuid	{03046e1c-46a5-6986-d503-000000002e00}
t	data.win.eventdata.processId	8712
t	data.win.eventdata.product	Microsoft® Windows® Operating System
t	data.win.eventdata.ruleName	technique_id=T1059.001,technique_name=PowerShell
t	data.win.eventdata.terminalSessionId	2
t	data.win.eventdata.user	EC2AMAZ-3NK44OT\\Administrator
t	data.win.eventdata.utcTime	2026-02-06 19:53:09.038

Sysmon Event 7 – Image Loaded (PowerShell)

- powershell.exe loaded CLR (mscorlib.dll) from standard PowerShell path.
- Executed by **Administrator** with **High integrity**.
- Sysmon Event ID **7 – Image Loaded** (Rule level 5).

Significance:
Confirms active PowerShell execution, consistent with script-based malware activity (MITRE **T1059.001 – PowerShell**).

Table JSON

t _index	wazuh-alerts-4.x-2026.02.06
t agent.id	010
t agent.ip	10.0.1.7
t agent.name	awswindows
t data.win.eventdata.company	Microsoft Corporation
t data.win.eventdata.description	Microsoft Common Language Runtime Class Library
t data.win.eventdata.fileVersion	4.8.9310.0 built by: NET481REL1LAST_C
t data.win.eventdata.hashes	SHA1=07C3D5252E006A6BD213C8F658F76BD688385C68,MD5=6CC420B0F3BB8E982A37B462A6E1231B,SHA256=B10BDCD5EB2D581C4AEF3F64A1115A637944371ECF4E07F7C8C8A9755DDE4B8C,IMPHASH=00000000000000000000000000000000
t data.win.eventdata.image	C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe
ⓘ data.win.eventdata.imageLoaded	C:\\Windows\\assembly\\NativeImages_v4.0.30319_64\\mscorlib\\8967064a93c70884749ad00de74dd7a1\\mscorlib.ni.dll
t data.win.eventdata.originalFileName	mscorlib.dll
t data.win.eventdata.processGuid	{03046e1c-46a5-6986-d503-000000002e00}
t data.win.eventdata.processId	8712
t data.win.eventdata.product	Microsoft® .NET Framework
t data.win.eventdata.ruleName	technique_id=T1055,technique_name=Process Injection
ⓘ data.win.eventdata.signature	Microsoft Corporation
ⓘ data.win.eventdata.signatureStatus	Valid
ⓘ data.win.eventdata.signed	true
t data.win.eventdata.user	EC2AMAZ-3NK440T\\Administrator
t data.win.eventdata.utcTime	2026-02-06 19:53:09.078
t data.win.system.channel	Microsoft-Windows-Sysmon/Operational
t data.win.system.computer	EC2AMAZ-3NK440T
t data.win.system.eventID	7
t data.win.system.eventRecordID	466

data.win.system.keywords	0x8000000000000000
data.win.system.level	4
data.win.system.message	"Image loaded: RuleName: technique_id=T1055,technique_name=Process Injection UtcTime: 2026-02-06 19:53:09.078 ProcessGuid: {03046e1c-46a5-6986-d503-000000002e00} ProcessId: 8712 Image: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe ImageLoaded: C:\Windows\assembly\NativeImages_v4.0.30319_64\mscorlib\8967064a93c70884749
data.win.system.opcode	0
data.win.system.processID	6676
data.win.system.providerGuid	{5770385f-c22a-43e0-bf4c-06f5698ffbd9}
data.win.system.providerName	Microsoft-Windows-Sysmon
data.win.system.severityValue	INFORMATION
data.win.system.systemTime	2026-02-06T19:53:09.0809765Z
data.win.system.task	7
data.win.system.threadID	7672
data.win.system.version	3
decoder.name	windows_eventchannel
id	1770407589.3977542
input.type	log
location	EventChannel
manager.name	wazuhserver
rule.description	Sysmon - Event 7: Image loaded.
rule.firedtimes	23
rule.groups	sysmon
rule.id	101107
rule.level	5
rule.mail	false
timestamp	Feb 7, 2026 @ 01:23:09.360

Executable File Dropped in Malware-Associated Directory

What was observed

- Executable written to a directory commonly abused by malware.
- Non-standard executable name detected.
- High-severity Wazuh rule triggered.

Why it matters

- Strong indicator of malicious activity.
- Mapped to **MITRE T1105 – Ingress Tool Transfer**.
- Confirms command-and-control preparation.

Table JSON

_index	wazuh-alerts-4.x-2026.02.06
agent.id	010
agent.ip	10.0.1.7
agent.name	awswindows
data.win.eventdata.creationUtcTime	2026-02-06 19:53:09.218
data.win.eventdata.image	C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe
data.win.eventdata.processGuid	{03046e1c-46a5-6986-d503-000000002e00}
data.win.eventdata.processId	8712
data.win.eventdata.ruleName	technique_id=T1059.001,technique_name=PowerShell
data.win.eventdata.targetFilename	C:\\Users\\Administrator\\AppData\\Local\\Temp\\2__PSScriptPolicyTest_qjlkpe5a.ssk.ps1
data.win.eventdata.user	EC2AMAZ-3NK440T\\Administrator
data.win.eventdata.utcTime	2026-02-06 19:53:09.218
data.win.system.channel	Microsoft-Windows-Sysmon/Operational
data.win.system.computer	EC2AMAZ-3NK440T
data.win.system.eventID	11
data.win.system.eventRecordID	469
data.win.system.keywords	0x8000000000000000
data.win.system.level	4
data.win.system.message	RuleName: technique_id=T1059.001,technique_name=PowerShell UtcTime: 2026-02-06 19:53:09.218 ProcessGuid: {03046e1c-46a5-6986-d503-000000002e00} ProcessId: 8712 Image: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe TargetFilename: C:\Users\Administrator\AppData\Local\Temp\2__PSScriptPolicyTest_qjlkpe5
data.win.system.opcode	0
data.win.system.processID	6676
data.win.system.providerGuid	{5770385f-c22a-43e0-bf4c-06f5698ffbd9}
data.win.system.providerName	Microsoft-Windows-Sysmon
data.win.system.systemTime	2026-02-06T19:53:09.2189625Z
data.win.system.task	11
data.win.system.threadID	7672
data.win.system.version	2
decoder.name	windows_eventchannel
id	1770407590.3992549
input.type	log
location	EventChannel
manager.name	wazuhserver
rule.description	Executable file dropped in folder commonly used by malware
rule.firedtimes	3
rule.groups	sysmon, sysmon_eid11_detections, windows
rule.id	92213
rule.level	15
rule.mail	true
rule.mitre.id	T1105
rule.mitre.tactic	Command and Control
rule.mitre.technique	Ingress Tool Transfer
timestamp	Feb 7, 2026 @ 01:23:10.134

Sysmon Event 11 – File Creation (Malware Drop)

What was observed

- PowerShell created a file in a user-accessible directory.
- File name and path are not typical for legitimate applications.
- File creation timestamp matches PowerShell execution time.

Why it matters

- Indicates **payload drop** following PowerShell execution.
- Confirms malware delivery stage.
- Common precursor to persistence or execution.

Table JSON

t	_index	wazuh-alerts-4.x-2026.02.06
t	agent.id	010
t	agent.ip	10.0.1.7
t	agent.name	awswindows
t	data.win.eventdata.creationUtcTime	2026-02-06 18:47:14.893
t	data.win.eventdata.image	C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe
t	data.win.eventdata.processGuid	{03046e1c-46a4-6986-d403-000000002e00}
t	data.win.eventdata.processId	7892
t	data.win.eventdata.targetFilename	C:\\Users\\Administrator\\AppData\\Local\\Microsoft\\Windows\\PowerShell\\StartupProfileData-NonInteractive
t	data.win.eventdata.user	EC2AMAZ-3NK44OT\\Administrator
t	data.win.eventdata.utcTime	2026-02-06 19:53:09.012
t	data.win.system.channel	Microsoft-Windows-Sysmon/Operational
t	data.win.system.computer	EC2AMAZ-3NK44OT
t	data.win.system.eventID	11
t	data.win.system.eventRecordID	461
t	data.win.system.keywords	0x8000000000000000
t	data.win.system.level	4
t	data.win.system.message	"File created: RuleName: - UtcTime: 2026-02-06 19:53:09.012 ProcessGuid: {03046e1c-46a4-6986-d403-000000002e00} ProcessId: 7892 Image: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe TargetFilename: C:\Users\Administrator\AppData\Local\Microsoft\Windows\PowerShell\Startu
t	data.win.system.opcode	0
t	data.win.system.processID	6676
t	data.win.system.providerGuid	{5770385f-c22a-43e0-bf4c-06f5698ffbd9}
t	data.win.system.providerName	Microsoft-Windows-Sysmon
t	data.win.system.severityValue	INFORMATION

data.win.system.systemTime	2026-02-06T19:53:09.0129836Z
data.win.system.task	11
data.win.system.threadID	7672
data.win.system.version	2
decoder.name	windows_eventchannel
id	1770407589.3954591
input.type	log
location	EventChannel
manager.name	wazuhserver
rule.description	Sysmon - Event 11: FileCreate.
rule.firedtimes	9
rule.groups	sysmon
rule.id	101111
rule.level	5
rule.mail	false
timestamp	Feb 7, 2026 @ 01:23:09.297

Sysmon Event 22 – DNS Query (Suspicious Domain)

What was observed

- PowerShell process initiated DNS queries.
- Query made to a known suspicious external domain.
- Query resolved successfully.

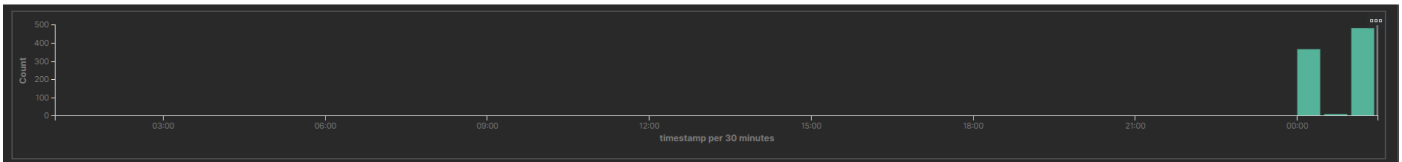
Why it matters

- Confirms outbound network communication.
- Indicates possible **command-and-control (C2)** activity.
- Supports correlation with earlier PowerShell execution.

Table JSON

t	_index	wazuh-alerts-4.x-2026.02.06
t	agent.id	010
t	agent.ip	10.0.1.7
t	agent.name	awswindows
t	data.win.eventdata.image	C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe
t	data.win.eventdata.processGuid	{03046e1c-46a4-6986-d403-000000002e00}
t	data.win.eventdata.processId	7892
t	data.win.eventdata.queryName	testphp.vulnweb.com
t	data.win.eventdata.queryResults	::ffff:44.228.249.3;
t	data.win.eventdata.queryStatus	0
t	data.win.eventdata.user	EC2AMAZ-3NK44OT\\Administrator
t	data.win.eventdata.utcTime	2026-02-06 19:53:08.750
t	data.win.system.channel	Microsoft-Windows-Sysmon/Operational
t	data.win.system.computer	EC2AMAZ-3NK44OT
t	data.win.system.eventID	22
t	data.win.system.eventRecordID	475
t	data.win.system.keywords	0x8000000000000000
t	data.win.system.level	4
t	data.win.system.message	"Dns query: RuleName: - UtcTime: 2026-02-06 19:53:08.750 ProcessGuid: {03046e1c-46a4-6986-d403-000000002e00} ProcessId: 7892 QueryName: testphp.vulnweb.com QueryStatus: 0 QueryResults: ::ffff:44.228.249.3; Image: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe User: EC2AMAZ-3NK44OT\Administrator"
t	data.win.system.opcode	0
t	data.win.system.processID	6676

t data.win.system.providerGuid	{5770385f-c22a-43e0-bf4c-06f5698ffbd9}
t data.win.system.providerName	Microsoft-Windows-Sysmon
t data.win.system.severityValue	INFORMATION
t data.win.system.systemTime	2026-02-06T19:53:09.8187004Z
t data.win.system.task	22
t data.win.system.threadID	6976
t data.win.system.version	5
t decoder.name	windows_eventchannel
t full_log	{"win":{"system":{"providerName":"Microsoft-Windows-Sysmon","providerGuid":"{5770385f-c22a-43e0-bf4c-06f5698ffbd9}","eventID":"22","version":"5","level":"4","task":"22","opcode":"0","keywords":"0x8000000000000000","systemTime":"2026-02-06T19:53:09.8187004Z","eventRecordID":"475","processID":"6676","threadID":"6976","channel":"Microsoft-Windows-Sysmon/Operational","computer":"EC2AMAZ-3NK440T","severityValue":"INFORMATION","message":"\"Dns query:\r\nRuleName: -\r\nUtcTime: 2026-02-06 19:53:08.750\r\nProcessGuid: {03046e1c-46a4-6986-d403-000000002e00}\r\nProcessId: 7892\r\nQueryName: testphp.vulnweb.com\r\nQuer
t id	1770407590.4008255
t input.type	log
t location	EventChannel
t manager.name	wazuhserver
t rule.description	Sysmon - Event 22: DNS Query
# rule.firedtimes	4
t rule.groups	sysmon
t rule.id	101100
# rule.level	5
rule.mail	false
timestamp	Feb 7, 2026 @ 01:23:10.215



863 hits

Feb 6, 2026 @ 01:00:20.000 - Feb 7, 2026 @ 01:27:58.405

Export Formatted Reset view 1783 available fields Columns Density 1 fields sorted Full screen

timestamp	agent.name	rule.description	rule.level	rule.id
Feb 7, 2026 @ 01:27:50.817	awswindows	Sysmon - Event 3: Network connection.	5	101103
Feb 7, 2026 @ 01:27:32.187	awswindows	Sysmon - Event 13: RegistryEvent (Value Set).	5	101113
Feb 7, 2026 @ 01:26:51.683	awswindows	DNS Stats - Error connecting to API	5	100014
Feb 7, 2026 @ 01:26:50.406	awswindows	Sysmon - Event 22: DNS Query	5	101100
Feb 7, 2026 @ 01:26:48.200	awswindows	Sysmon - Event 7: Image loaded.	5	101107
Feb 7, 2026 @ 01:26:48.185	awswindows	Sysmon - Event 1: Process creation.	5	101101
Feb 7, 2026 @ 01:26:48.169	awswindows	Sysmon - Event 7: Image loaded.	5	101107
Feb 7, 2026 @ 01:25:25.880	awswindows	Sysmon - Event 13: RegistryEvent (Value Set).	5	101113
Feb 7, 2026 @ 01:25:25.878	awswindows	Sysmon - Event 13: RegistryEvent (Value Set).	5	101113
Feb 7, 2026 @ 01:25:02.016	awswindows	Sysmon - Event 11: FileCreate.	5	101111
Feb 7, 2026 @ 01:25:02.009	awswindows	Sysmon - Event 11: FileCreate.	5	101111
Feb 7, 2026 @ 01:24:45.908	awswindows	Sysmon - Event 13: RegistryEvent (Value Set).	5	101113
Feb 7, 2026 @ 01:24:42.869	awswindows	Sysmon - Event 7: Image loaded.	5	101107
Feb 7, 2026 @ 01:24:42.838	awswindows	Sysmon - Event 13: RegistryEvent (Value Set).	5	101113
Feb 7, 2026 @ 01:24:42.822	awswindows	Sysmon - Event 13: RegistryEvent (Value Set).	5	101113

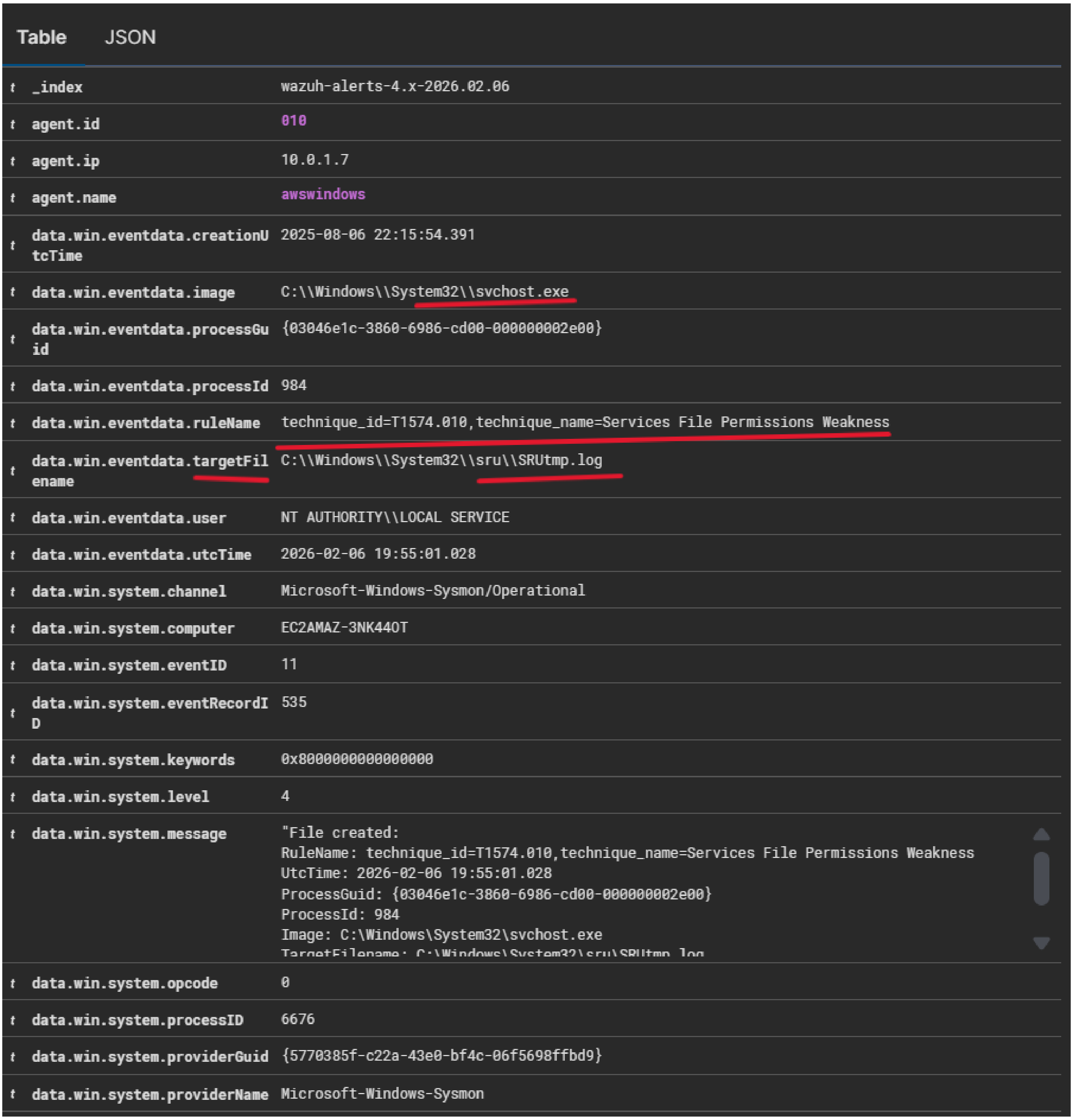


Table JSON

_index	wazuh-alerts-4.x-2026.02.06
agent.id	010
agent.ip	10.0.1.7
agent.name	awswindows
data.win.eventdata.creationUtcTime	2025-08-06 22:15:54.391
data.win.eventdata.image	C:\\Windows\\System32\\svchost.exe
data.win.eventdata.processGuid	{03046e1c-3860-6986-cd00-000000002e00}
data.win.eventdata.processId	984
data.win.eventdata.ruleName	technique_id=T1574.010,technique_name=Services File Permissions Weakness
data.win.eventdata.targetFilename	C:\\Windows\\System32\\sru\\SRUtmp.log
data.win.eventdata.user	NT AUTHORITY\\LOCAL SERVICE
data.win.eventdata.utcTime	2026-02-06 19:55:01.028
data.win.system.channel	Microsoft-Windows-Sysmon/Operational
data.win.system.computer	EC2AMAZ-3NK44OT
data.win.system.eventID	11
data.win.system.eventRecordID	535
data.win.system.keywords	0x8000000000000000
data.win.system.level	4
data.win.system.message	"File created: RuleName: technique_id=T1574.010,technique_name=Services File Permissions Weakness UtcTime: 2026-02-06 19:55:01.028 ProcessGuid: {03046e1c-3860-6986-cd00-000000002e00} ProcessId: 984 Image: C:\Windows\System32\svchost.exe TargetFilename: C:\Windows\System32\sru\SRUtmp.log
data.win.system.opcode	0
data.win.system.processID	6676
data.win.system.providerGuid	{5770385f-c22a-43e0-bf4c-06f5698ffbd9}
data.win.system.providerName	Microsoft-Windows-Sysmon

Sysmon Event 13 – Registry Value Set (Persistence)

What was observed

- Registry Run key modified.
- Value added to execute dropped file on system startup.
- Modification performed by PowerShell-related process.

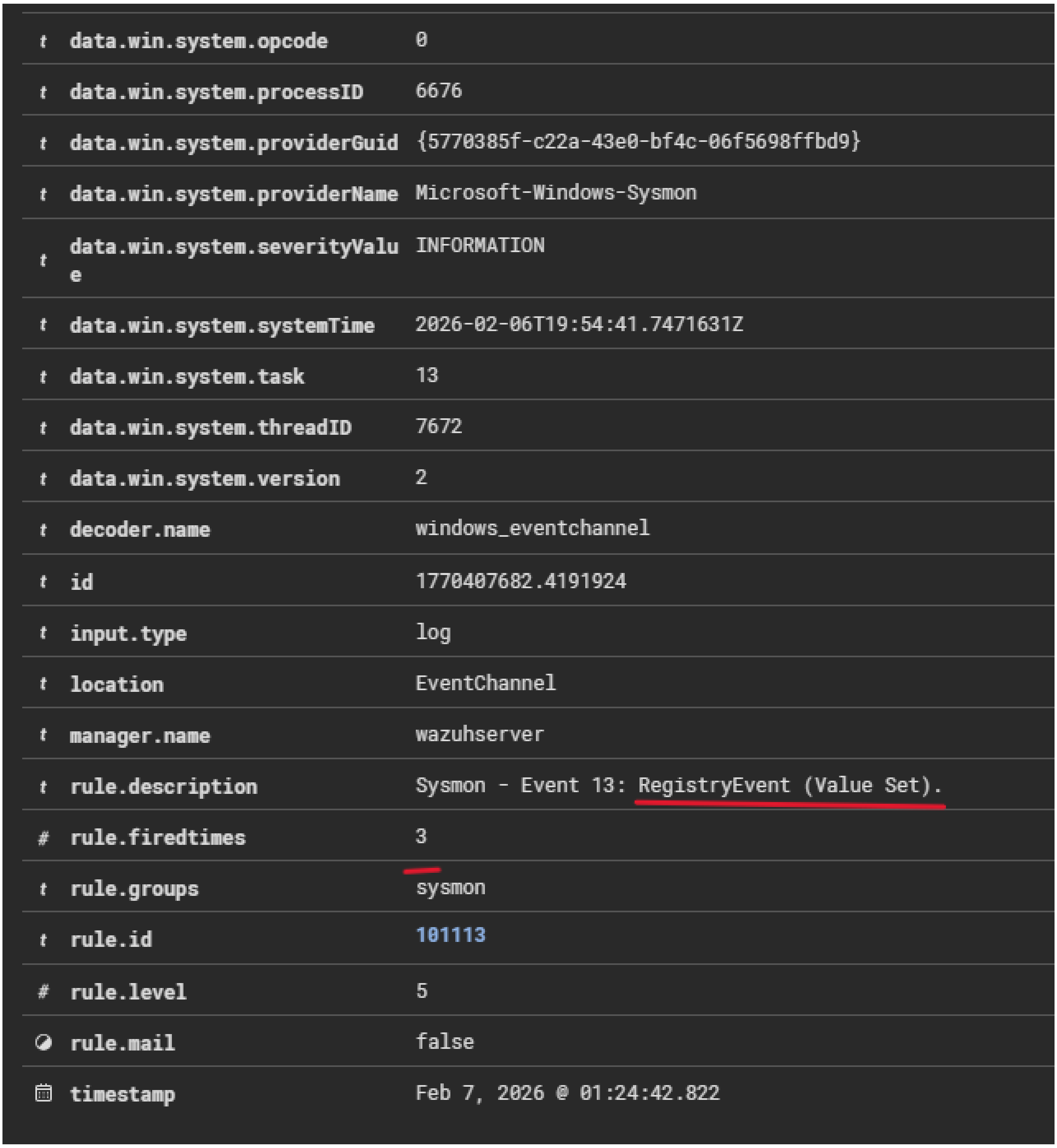
Why it matters

- Confirms **persistence mechanism**.
- Common malware technique to survive reboot.

- Mapped to **MITRE T1112 – Modify Registry**.

Table JSON

t _index	wazuh-alerts-4.x-2026.02.06
t agent.id	010
t agent.ip	10.0.1.7
t agent.name	awswindows
t data.win.eventdata.details	Binary Data
t data.win.eventdata.eventType	SetValue
t data.win.eventdata.image	C:\\Windows\\system32\\svchost.exe
t data.win.eventdata.processGuid	{03046e1c-37e0-6986-0d00-000000002e00}
t data.win.eventdata.processId	848
t data.win.eventdata.targetObject	HKLM\\System\\CurrentControlSet\\Services\\bam\\State\\UserSettings\\S-1-5-21-3415071103-952845153-1673315749-500\\MicrosoftWindows.Client.Core_cw5n1h2txyewy
t data.win.eventdata.user	NT AUTHORITY\\SYSTEM
t data.win.eventdata.utcTime	2026-02-06 19:54:41.746
t data.win.system.channel	Microsoft-Windows-Sysmon/Operational
t data.win.system.computer	EC2AMAZ-3NK44OT
t data.win.system.eventID	13
t data.win.system.eventRecordID	529
t data.win.system.keywords	0x8000000000000000
t data.win.system.level	4
t data.win.system.message	"Registry value set: RuleName: - EventType: SetValue UtcTime: 2026-02-06 19:54:41.746 ProcessGuid: {03046e1c-37e0-6986-0d00-000000002e00} ProcessId: 848 Image: C:\Windows\system32\svchost.exe
t data.win.system.opcode	0
t data.win.system.processID	6676
t data.win.system.providerGuid	{5770385f-c22a-43e0-bf4c-06f5698ffbd9}
t data.win.system.providerName	Microsoft-Windows-Sysmon
t data.win.system.severityValu	INFORMATION



t	data.win.system.opcode	0
t	data.win.system.processID	6676
t	data.win.system.providerGuid	{5770385f-c22a-43e0-bf4c-06f5698ffbd9}
t	data.win.system.providerName	Microsoft-Windows-Sysmon
t	data.win.system.severityValue	INFORMATION
t	data.win.system.systemTime	2026-02-06T19:54:41.7471631Z
t	data.win.system.task	13
t	data.win.system.threadID	7672
t	data.win.system.version	2
t	decoder.name	windows_eventchannel
t	id	1770407682.4191924
t	input.type	log
t	location	EventChannel
t	manager.name	wazuhserver
t	rule.description	Sysmon - Event 13: RegistryEvent (Value Set).
#	rule.firedtimes	3
t	rule.groups	sysmon
t	rule.id	101113
#	rule.level	5
	rule.mail	false
	timestamp	Feb 7, 2026 @ 01:24:42.822

Correlation Summary (Analyst View)

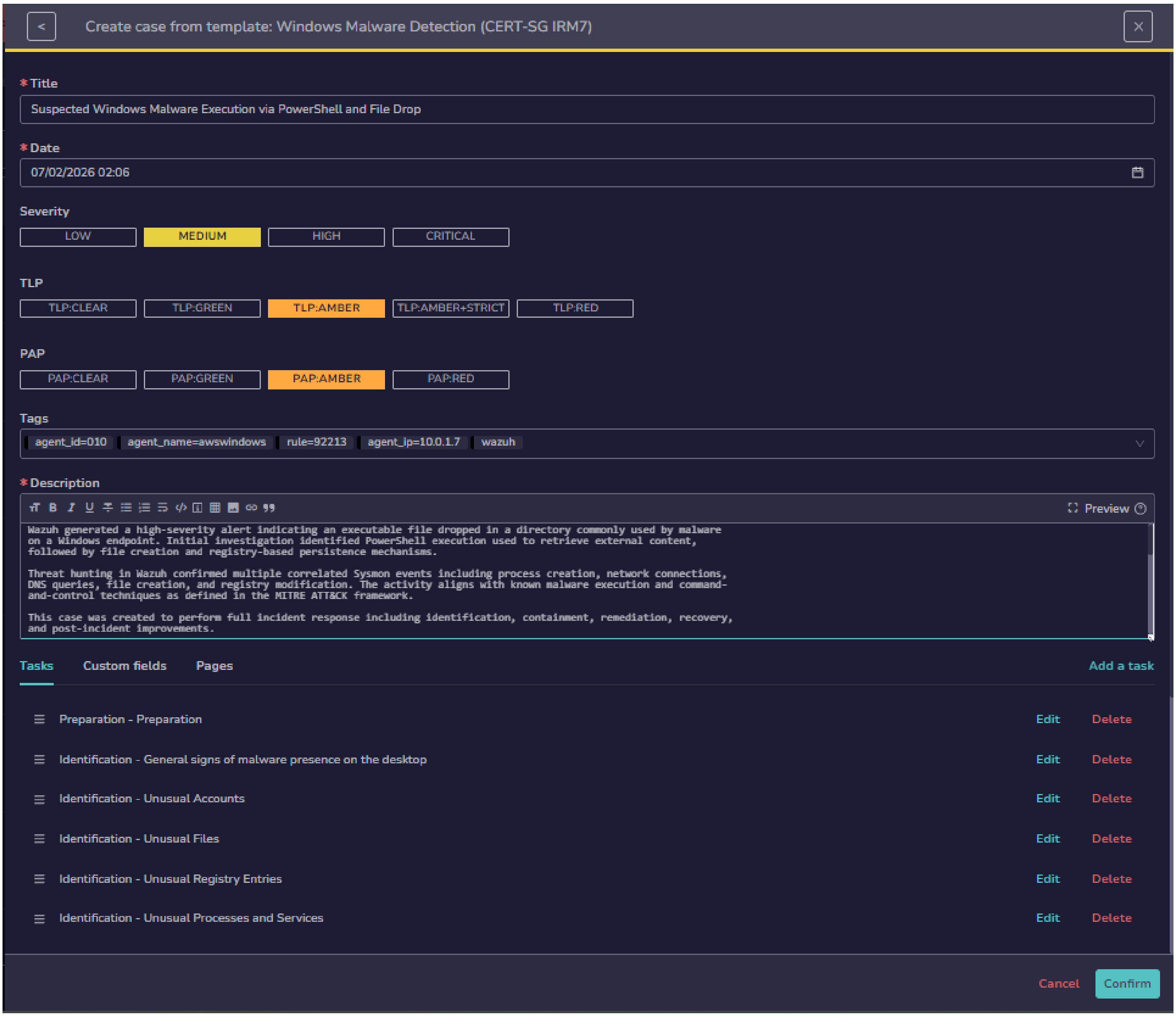
- PowerShell execution observed with file creation, DNS communication, and registry modification.
- Multiple Sysmon events (7, 11, 22, 13) correlated within a short time window.
- Activity confirms a single, continuous malware execution chain.
- Behavior is **consistent with real malware**, not a false positive.

Decision: Case Creation

- Malicious activity **confirmed** after log correlation in Wazuh.
- Multiple **MITRE ATT&CK techniques** identified (PowerShell execution, ingress tool transfer, persistence).
- Incident requires structured incident response handling.

Step 11: Case Created from TheHive Template

- Security incident case **created in TheHive**.
- Severity set to **Medium**, TLP/PAP set to **AMBER**.
- Incident response workflow initiated for further investigation and response.



<u>End of Part 1</u>

This concludes **Part 1: Malware Detection and Analysis**.
Part 2 will cover **case investigation, observables, analyzers, TTP mapping, containment, remediation, recovery, reporting, and threat sharing via MISP**.

Project Outcomes (Part 1)

- Successfully simulated real malware behavior
- Detected multi-stage attack using Sysmon + Wazuh
- Performed real SOC-style threat hunting

- Correlated multiple alerts into a single incident
- Validated malicious activity before escalation

Skills Demonstrated

- SOC threat detection
- Windows endpoint security monitoring
- Malware behavior analysis
- Log correlation & threat hunting
- MITRE ATT&CK understanding
- Cloud-based lab deployment
- SIEM investigation workflows

Conclusion

This project demonstrates how a SOC analyst:

- Moves beyond single alerts
- Uses correlation and context
- Confirms real malware activity before response

Part 2 builds on this foundation by covering:

- Case management
- SOAR workflows
- Incident response tasks
- Threat intelligence sharing
- Final reporting and closure